

Appendix B

Acquiring Software

Caveat receptor . . .

Anything free comes with no guarantee!

— JACK DONGARRA and ERIC GROSSE, *Netlib mail header*

In this appendix we provide information on how to acquire software mentioned in the book. First, we describe some basic aspects of the Internet.

B.1. Internet

A huge variety of information and software is available over the Internet, the worldwide combination of interconnected computer networks. The location of a particular object is specified by a URL, which stands for “Uniform Resource Locator”. Examples of URLs are

```
http://www.netlib.org/index.html  
ftp://ftp.netlib.org
```

The first example specifies a World Wide Web server (http = hypertext transfer protocol) together with a file in hypertext format (html = hypertext markup language), while the second specifies an anonymous ftp site. In any URL, the site address may, optionally, be followed by a filename that specifies a particular file. For more details about the Internet see on-line information, or one of the many books on the subject.

B.2. Netlib

Netlib is a repository of freely available mathematical software, documents, and databases of interest to the scientific computing community [350, 1987], [168, 1994]. It includes

- research codes,
- golden oldies (classic programs that are not available in standard libraries),
- the collected algorithms of the ACM,
- program libraries such as EISPACK, LINPACK, LAPACK, and MINPACK.
- back issues of NA-Digest, a weekly digest for the numerical analysis community,
- databases of conferences and performance data for a wide variety of machines.

Netlib also enables the user to download technical reports from certain institutions, to download software and errata for textbooks, and to search a “white pages” database.

Netlib can be accessed in several ways.

- Over the World Wide Web: `http://www.netlib.org`
- By anonymous ftp: `ftp://ftp.netlib.org`

- By electronic mail. For an introduction and master index, send a one-line email message as follows:
`mail netlib@netlib.org`
`send index`

Netlib is mirrored at various sites throughout the world.

B.3. MATLAB

MATLAB is a commercial program sold by The MathWorks, Inc. It runs on a variety of platforms. The MathWorks maintains a collection of user-contributed M-files, which is accessible over the Internet.

For information contact

The MathWorks, Inc.
3 Apple Hill Drive
Natick, MA 01760-2098
USA

Tel: +508-647-7000
Fax: +508-647-7001

Web: www.mathworks.com
Newsgroup: comp.soft-sys.matlab
FTP: [ftp.mathworks.com](ftp://ftp.mathworks.com)
E-mail: info@mathworks.com

B.4. NAG Library and NAGWare F95 Compiler

The Numerical Algorithms Group (NAG) produces a variety of software products. Relevant to this book are the f95 compiler and the NAG Library, a large numerical program library available in several programming languages.

For information contact

NAG Ltd.
Wilkinson House
Jordan Hill Road
Oxford, OX2 8DR
UK

Tel: +44 1865 511245
Fax: +44 1865 310139

email: sales@nag.co.uk
Web: <http://www.nag.co.uk>

NAG has subsidiaries and distributors, whose addresses can be obtained from the above sources.